

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications

Course Title: **Probability & Statistics**

Code No: Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

11. Describe the scope and limitations of Statistics.
12. What do you mean by statistics? The following table represents the marks of Probability and Statistics of 100 students.
13. Define correlation. From the following data on marks of 10 students in the two subjects, calculate the Karl Pearson's coefficient of correlation and interpret the result:
14. Define regression. The following table gives the age of the computers of a certain company and annual maintenance costs:
15. Define Poisson distribution. In certain factory timing out optical lenses, there is a small chance, $1/500$ for any lens to be defective. The lenses are supplied in a packet of 10 each. What is the probability that a packet will contain: (i) No defective lens, (ii) At least one defective lenses, (iii) At most two defective lenses.
16. A dean of a college wants to use the mean of a random sample to estimate the average amount of time students take to get from one class to the next, and she wants to be able to assert with 95% confidence that error is at most 0.25 minute. If it can be presumed from experience that $\sigma = 1.40$ minutes, how large a sample will she have to take?
17. Define sampling. A population consists of the four numbers 2, 8, 14 and 20. (i) Write down all possible sample size of two without replacement. (ii) Verify that the sample mean is an unbiased estimate of population mean.

Group C

Attempt any TWO questions [2x10=20]

18. Two computer manufacturers A and B compete for profitable and prestigious contract. In their rivalry, each claim that their computer is consistent. For this it was decided to start execution of the same program simultaneously on 50 computers of each company and recorded the time as given below.
19. Define normal distribution and standard normal distribution. The life time of a certain electronic component is a normal distribution with mean 5000 hours and standard deviation of 1000 hours, compute the probabilities

under the following conditions:(a) Life time of components is less than 5012 hours,(b) Lifetime of components between 4000 to 6000 hours, and,(c) Lifetime of components more than 7000 hours.

- 20.** Write the properties of design of experiment. The lifetime in hours of samples from three different brands of batteries were recorded with the following results: